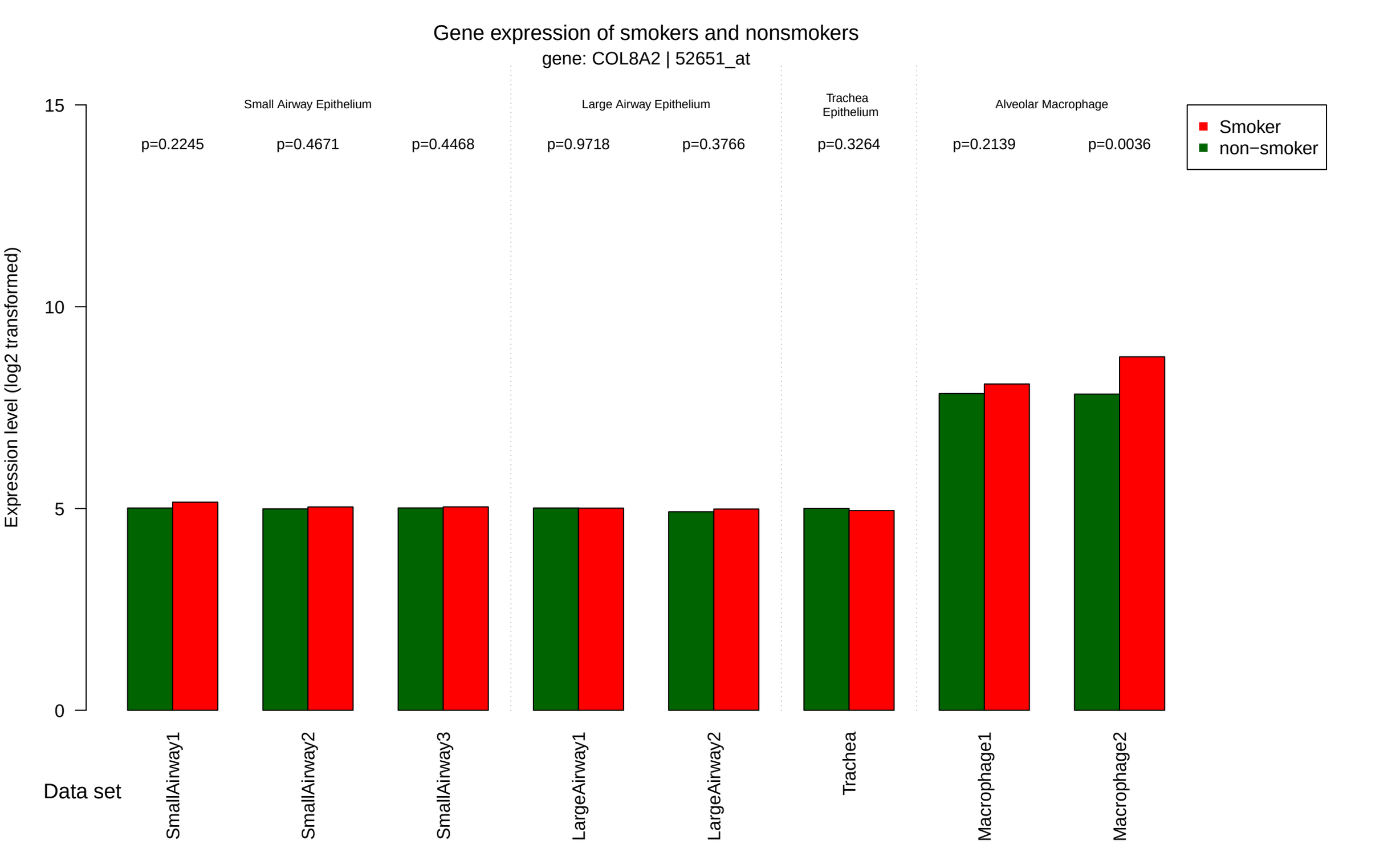
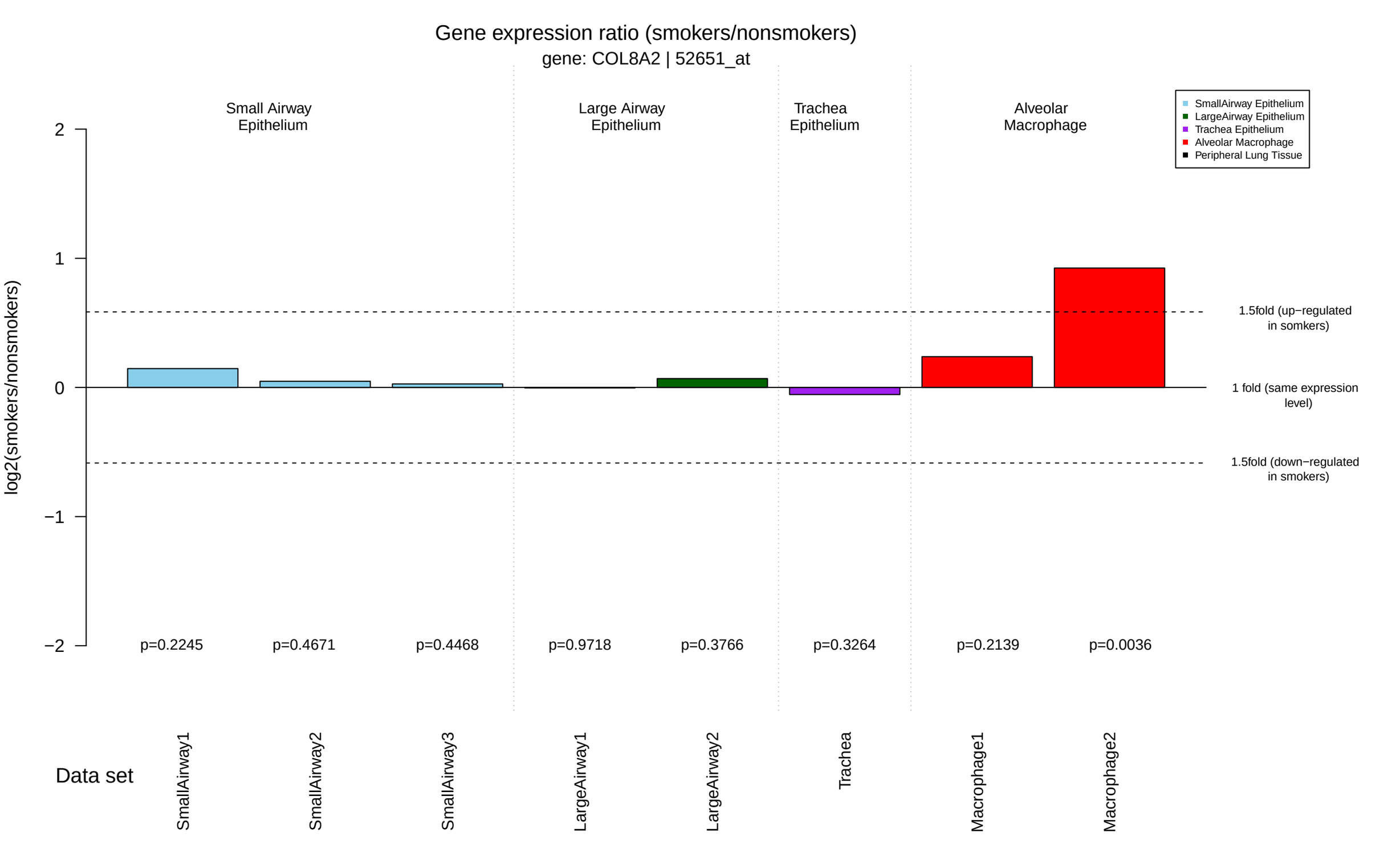
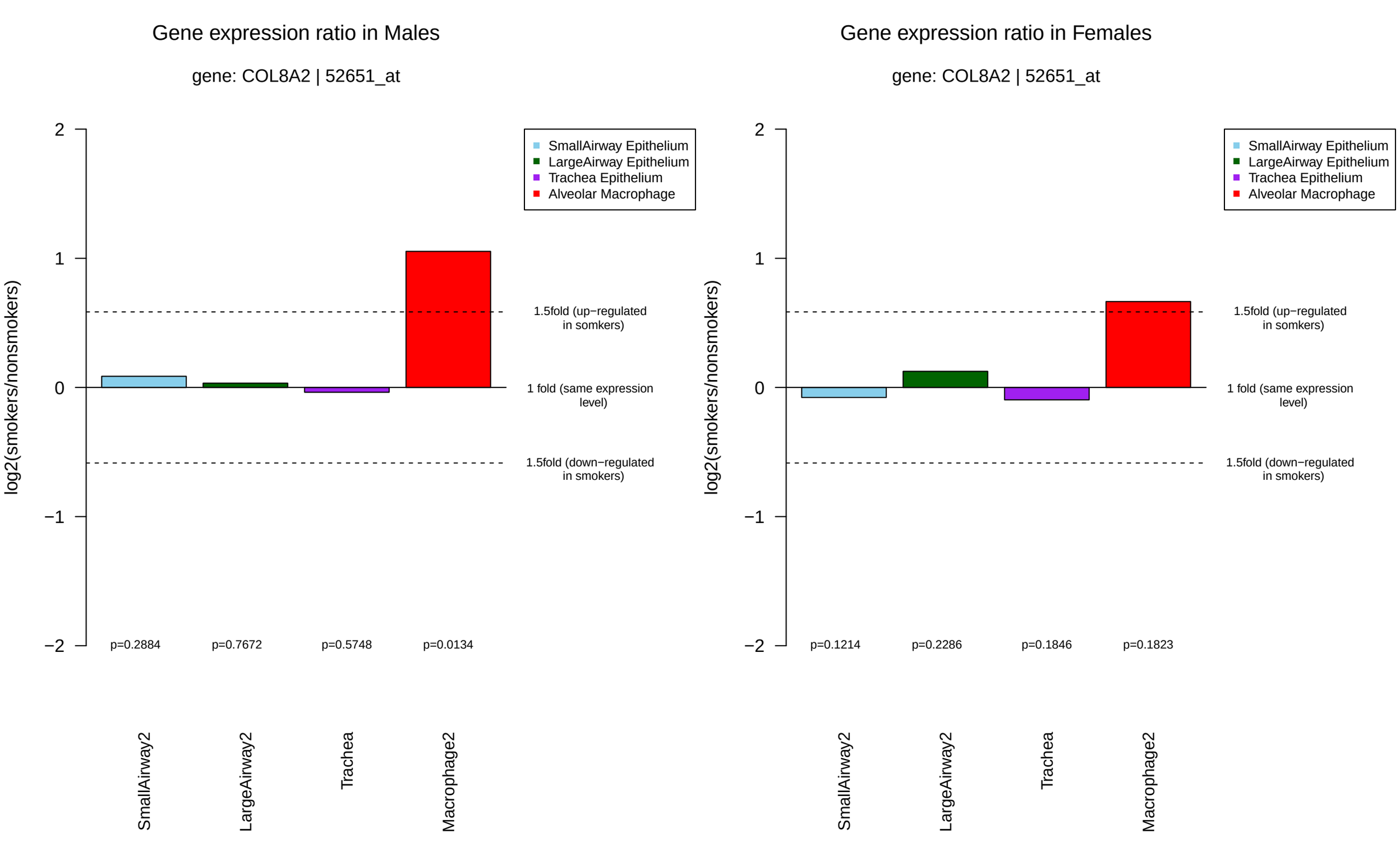
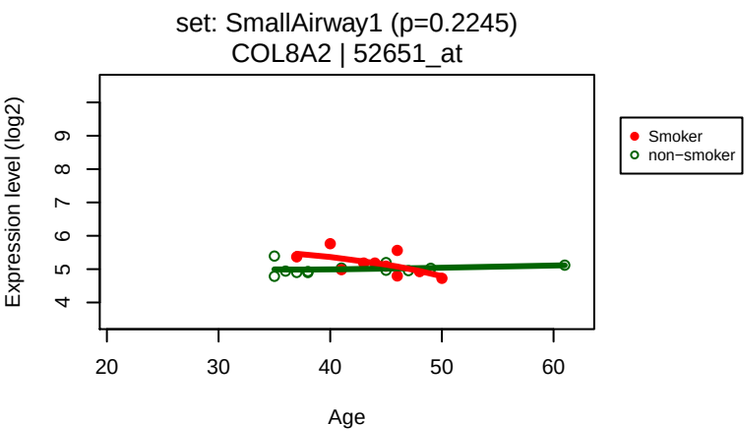
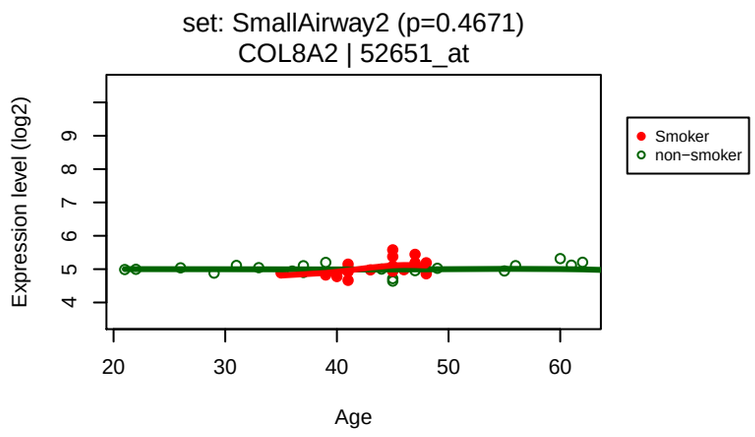
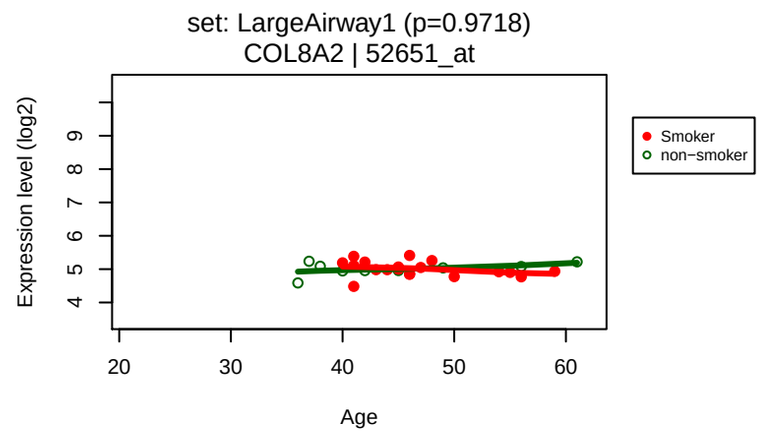

Gene expression of smokers and nonsmokers
gene: COL8A2 | 52651_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2245
p=0.4671
p=0.4468
p=0.9718
p=0.3766
p=0.3264
p=0.2139
p=0.0036
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

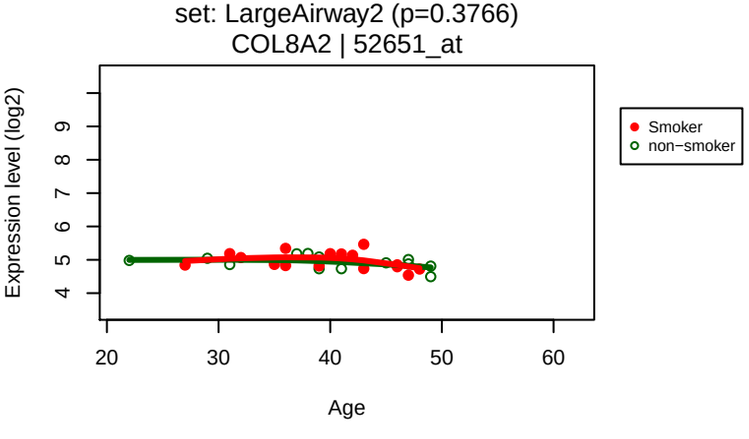

Gene expression ratio (smokers/nonsmokers)
gene: COL8A2 | 52651_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2245
p=0.4671
p=0.4468
p=0.9718
p=0.3766
p=0.3264
p=0.2139
p=0.0036
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

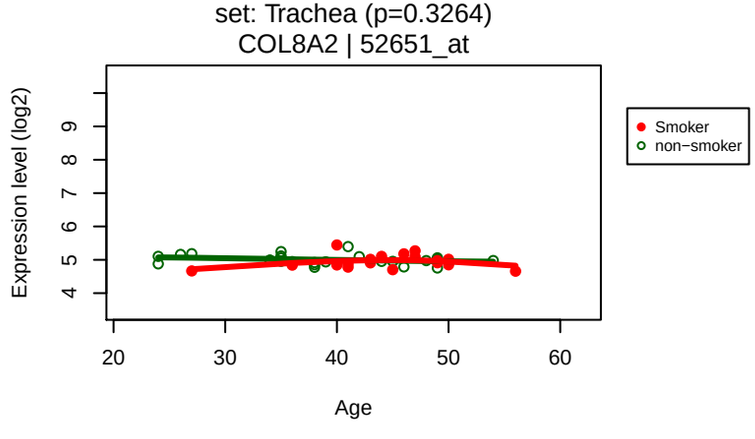

Gene expression ratio in Males
gene: COL8A2 | 52651_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2884
p=0.7672
p=0.5748
p=0.0134
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL8A2 | 52651_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1214
p=0.2286
p=0.1846
p=0.1823
SmallAirway2
LargeAirway2
Trachea
Macrophage2

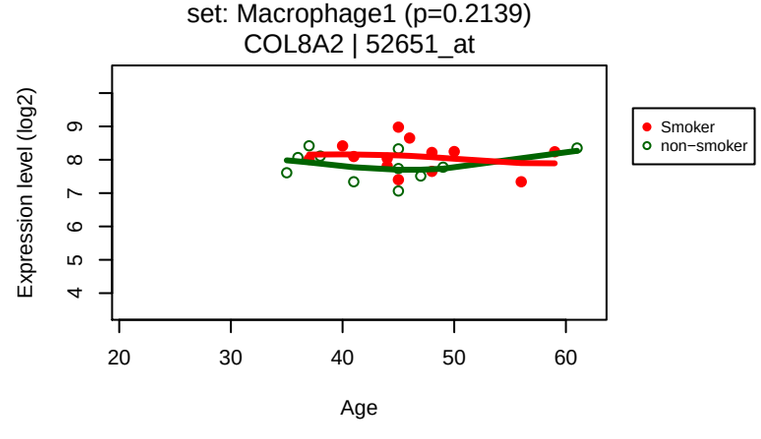

set: SmallAirway1 (p=0.2245)
COL8A2 | 52651_at
Expression level (log2)
Age
Smoker
non−smoker

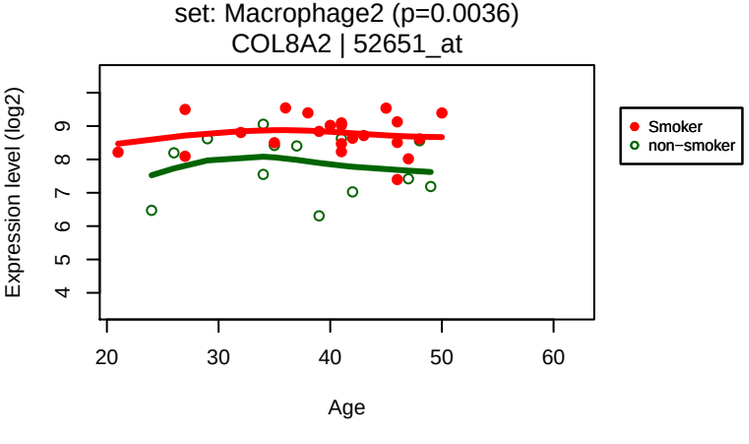

set: SmallAirway2 (p=0.4671)
COL8A2 | 52651_at
Expression level (log2)
Age
Smoker
non−smoker

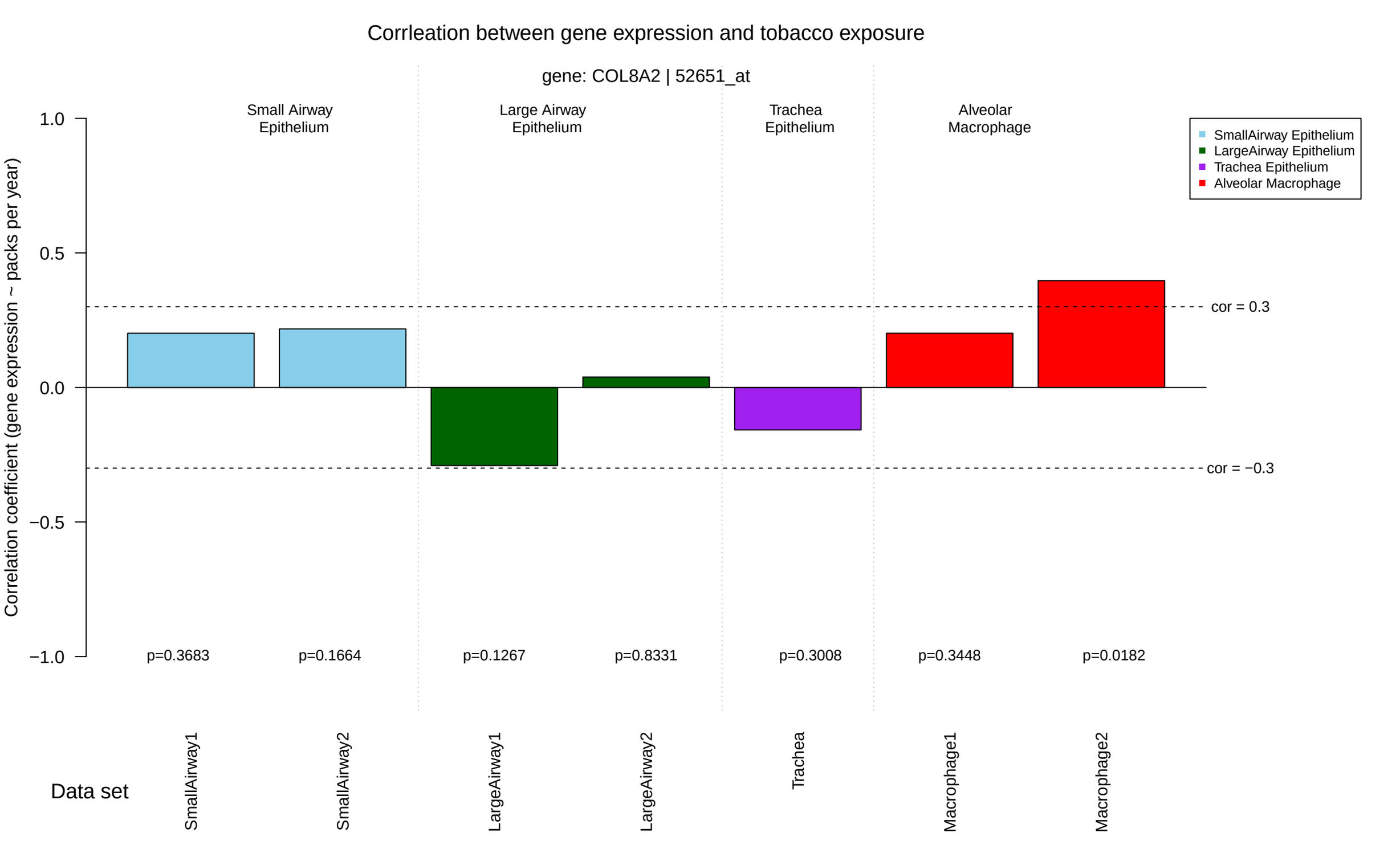

set: LargeAirway1 (p=0.9718)
COL8A2 | 52651_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3766)
COL8A2 | 52651_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3264)
COL8A2 | 52651_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2139)
COL8A2 | 52651_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0036)
COL8A2 | 52651_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL8A2 | 52651_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3683
p=0.1664
p=0.1267
p=0.8331
p=0.3008
p=0.3448
p=0.0182
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2